

Machine Learning for Solid-State Battery Materials Discovery

Abstract

The paradigm shift from conventional liquid-electrolyte lithium-ion batteries to all-solid-state batteries (SSBs) represents a critical frontier in advanced energy storage, promising substantial enhancements in volumetric and gravimetric energy density, operational safety, and thermal stability. However, the commercial realization of SSBs is systematically impeded by the arduous challenge of discovering high-performance solid-state electrolytes (SSEs) that concurrently exhibit superionic conductivity at room temperature, robust mechanical properties to suppress dendrite formation, and wide electrochemical stability windows. Traditional materials discovery paradigms, which rely heavily on chemical intuition, trial-and-error experimentation, and computationally exhaustive *ab initio* calculations such as density functional theory (DFT), are fundamentally ill-equipped to navigate the practically infinite combinatorial space of inorganic materials. Consequently, artificial intelligence and machine learning have catalyzed a transformative era in computational materials science.

This comprehensive survey systematically examines the integration of machine learning algorithms in the discovery, design, and optimization of solid-state battery materials. The analysis delineates the evolution of data-driven workflows, commencing with the curation of high-fidelity materials databases and the sophisticated engineering of structural and compositional descriptors. The survey scrutinizes high-throughput computational screening protocols, augmented by active learning and Bayesian optimization, which efficiently isolate superionic conductors from vast repositories of hypothetical candidates. Furthermore, the development of machine learning interatomic potentials (MLIPs)—including universal equivariant graph neural networks such as MACE, CHGNet, M3GNet, and ALIGNN—is evaluated for its capacity to extend atomistic simulations to unprecedented spatiotemporal scales with near-DFT accuracy, thereby enabling the simulation of dynamic ion transport and complex interfacial phenomena. The synthesis of generative artificial intelligence, notably the Crystal Diffusion Variational Autoencoder (CDVAE) and massive exploration frameworks like GNoME, demonstrates the nascent capability to perform *de novo* crystal structure prediction. Finally, the survey evaluates the translation of virtual discoveries into physical materials via autonomous robotic synthesis platforms, such as the A-Lab, and outlines the enduring challenges in modeling multivalent ion transport and resolving the electro-chemo-mechanical intricacies of solid-solid interfaces. By mapping these interconnected methodologies, this paper provides a strategic roadmap for the next generation of accelerated, closed-loop materials discovery.

Introduction

In the pursuit of sustainable, decarbonized energy solutions, the widespread electrification of transportation and the stabilization of grid-scale renewable energy systems depend

fundamentally on the advancement of electrochemical energy storage [1, 2]. Conventional lithium-ion batteries, which have dominated the market for decades, are rapidly approaching their theoretical limits regarding energy density. Furthermore, these systems are increasingly scrutinized for safety vulnerabilities stemming from their reliance on highly flammable organic liquid electrolytes [3, 4]. All-solid-state batteries (SSBs), which systematically replace liquid electrolytes with solid ion conductors, offer a transformative alternative. By utilizing inorganic ceramic, sulfide, halide, or polymeric solid-state electrolytes (SSEs), SSBs hold the potential to enable the use of metallic lithium anodes. The integration of lithium metal drastically increases specific energy while entirely eliminating the leakage and combustion risks associated with volatile liquids [5, 6].

Despite these profound theoretical advantages, the practical realization of SSBs is obstructed by severe material limitations. Ideal SSEs must fulfill a stringent and often conflicting set of multi-objective criteria. They require exceptionally high room-temperature ionic conductivity (approaching or exceeding 10^{-3} S/cm to match liquid counterparts), low migration energy barriers (E_a), broad electrochemical stability windows against highly reducing lithium anodes and strongly oxidizing high-voltage cathodes, and sufficient mechanical rigidity (specifically, high shear moduli) to physically suppress lithium dendrite propagation [7, 8]. Current candidate classes—including oxides (e.g., garnets like $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ and NASICON-type materials), sulfides (e.g., argyrodites like $\text{Li}_6\text{PS}_5\text{Cl}$), and emerging complex halides—often exhibit critical performance trade-offs. For instance, sulfide-based SSEs frequently boast high ionic conductivities due to highly polarizable sulfur frameworks but suffer from extreme moisture sensitivity and narrow electrochemical stability windows [2, 9, 40]. Conversely, oxide garnets offer excellent thermodynamic stability but require impractically high sintering temperatures, leading to poor interfacial contact [2, 10].

The traditional materials discovery pipeline relies on heuristic chemical intuition and sequential experimental synthesis, an inherently slow process that typically requires decades to transition a novel material from laboratory conceptualization to commercial deployment [11, 39].

Computational methods, primarily based on density functional theory (DFT) and *ab initio* molecular dynamics (AIMD), have partially accelerated this process by enabling the virtual calculation of activation energies, band gaps, and phase stability [3, 12, 40]. However, the scaling laws of quantum mechanical solvers severely restrict their application to small systems (typically hundreds of atoms) and highly constrained simulation timescales (picoseconds). Given the structural complexity of superionic conductors—which often feature partial atomic occupancies, mixed-anion sublattices, local configuration fluctuations, and highly disordered topological networks—DFT-driven exploration across the vast chemical space of millions of hypothetical stoichiometries is computationally intractable [5, 12, 13].

Machine learning has rapidly emerged as the definitive solution to bridge this vast spatiotemporal divide [4, 5, 8]. By extracting complex, non-linear structure-property relationships from high-fidelity theoretical and experimental data, predictive artificial intelligence algorithms bypass the need to explicitly solve the Schrödinger equation for every

candidate configuration [14, 15]. The integration of machine learning operates across multiple interconnected abstraction levels. Supervised regression models rapidly screen libraries for potential fast-ion conductors using basic structural inputs [9, 10]. Universal interatomic potentials emulate DFT accuracy for large-scale dynamic simulations, illuminating defect chemistry and interfacial evolution at the microsecond scale [16, 17]. Generative models, operating in continuous latent spaces, propose entirely novel crystal structures unrestricted by existing databases [36, 37]. This survey systematically unpacks the algorithmic architectures, curated databases, and closed-loop robotic systems powering this revolution, providing a comprehensive assessment of how artificial intelligence is decoding the multiscale complexities of solid-state battery materials.

Methods: Data Infrastructure and Feature Representation

The fundamental efficacy of any machine learning protocol is inextricably linked to the quality, scale, and specific mathematical representation of its underlying data. In solid-state materials science, the transition from discrete experimental observations to large-scale predictive modeling necessitates rigorous data infrastructure and sophisticated geometric encoding. The curation of extensive, diverse, and high-fidelity datasets is the prerequisite for training models that can generalize beyond their immediate training domains.

Materials Databases and Datasets

The foundation of modern machine learning in battery materials discovery is built upon high-throughput computational databases and curated experimental repositories. The Materials Project (MP) serves as the primary engine for theoretical data, offering computed formation energies, band gaps, elastic tensors, and relaxed atomic geometries derived via automated DFT workflows for over 150,000 inorganic compounds [10, 18, 30]. Complementary databases such as the Open Quantum Materials Database (OQMD) and AFLOWLIB expand the available phase space with additional thermodynamic and electronic structure data, while JARVIS-DFT extends properties to include phonons, elastic constants, and diffusion metrics [18, 38]. Experimental grounding is provided by the Inorganic Crystal Structure Database (ICSD), which archives hundreds of thousands of experimentally validated crystal structures, including precise lattice parameters and atomic coordinates necessary for verifying theoretical predictions [10, 18].

For the specific domain of solid-state electrolytes, predicting dynamic properties like ionic conductivity requires specialized datasets, as standard static databases often lack information on migration barriers and defect energetics. To address this, specialized repositories have been recently established. For instance, the Liverpool Ionics (Lilon) dataset and the OBELiX database compile hundreds to thousands of experimentally measured room-temperature ionic conductivities, linking them directly to their structural families and measured compositions [13, 15]. Similarly, the Dynamic Database of Solid-State Electrolyte (DDSE) captures conductivity measurements across broad temperature ranges, facilitating the extraction of pre-exponential factors and activation energies from Arrhenius relations [14]. Advanced dynamical training sets,

such as LiTraj, provide over 130,000 atomic configurations with energies and forces captured during Nudged Elastic Band (NEB) optimizations, which is an essential resource for training models to understand precise lithium diffusion pathways [14, 15].

Database / Dataset	Primary Focus	Origin Modality	Key Application in SSB ML
Materials Project (MP)	Computed thermodynamics, structures, band gaps	Theoretical (DFT)	Base pretraining for Graph Neural Networks; phase stability screening [10, 18].
ICSD	Validated crystal structures, lattice constants	Experimental	Structural verification; input for unsupervised structural learning [10, 18].
OBELiX / Lilon	Ionic conductivities, space groups	Experimental	Supervised learning targets for predicting macroscopic conductivity [13, 15].
LiTraj	Transition states, NEB trajectories, forces	Theoretical (DFT)	Training MLIPs to predict precise lithium migration barriers [14].
MatBench	ML-ready standardized benchmark datasets	Mixed	Algorithm benchmarking for formation energy and band gap prediction [18, 38].

Feature Engineering and Material Representation

Transforming a 3D periodic crystal into a machine-readable mathematical format requires descriptors that are invariant to translation, rotation, and arbitrary permutations of atomic

indexing. Early machine learning implementations relied heavily on composition-based descriptors (e.g., Magpie features), encoding scalar properties like average electronegativity, atomic radii, and valence electron counts directly from the chemical formula [7, 10, 11]. Recent innovations have demonstrated that composition-only frameworks, when augmented with sophisticated physics-based features such as ionic radius mismatch, can achieve highly robust correlations with ionic conductivity without requiring relaxed 3D coordinates [6, 7]. This structure-free approach circumvents a major computational bottleneck: the inability to rapidly screen hypothetical compositions whose exact crystal structures are not yet known. However, because ionic transport is deeply dependent on topological bottlenecks, polyhedral connectivity, and the free volume within a lattice, structural descriptors are generally superior for precision modeling. Representations based on Voronoi tessellations, Coulomb matrices, and local environment fingerprints explicitly capture the spatial distribution of atoms and their local coordination environments [10, 12, 14]. Advanced models construct Bond Valence Site Energy (BVSE) maps to proxy the electrostatic potential landscape, effectively transforming continuous 3D scalar fields into predictive, low-dimensional features for mapping migration barriers [10, 14].

Methods: High-Throughput Screening and Property Prediction

The identification of viable solid-state electrolytes historically relied on iterative elemental substitution within known superionic frameworks—for example, halogenating sulfide argyrodites to induce structural disorder and enhance lithium mobility [1, 2]. Machine learning reformulates this targeted approach into a high-throughput virtual screening (HTVS) task over vast combinatorial spaces, evaluating millions of hypothetical candidates instantaneously based on learned physical rules.

Supervised Property Classification and Regression

Using supervised learning, classification models differentiate between fast and slow ion conductors, while regression models predict specific scalar values such as the activation

energy (E_a) or room-temperature ionic conductivity (σ). Early pioneering work in the field employed logistic regression algorithms to filter over 12,000 lithium-containing solids [8, 9]. By predicting candidate electrolytes based on a balance of ionic conductivity probabilities, DFT-calculated thermodynamic stabilities, and electronic band gaps, researchers identified new families of superionic conductors, such as lithium-boron-sulfur (Li-B-S) systems, that possessed high conductivities and wide electrochemical windows [8].

Modern implementations utilize sophisticated ensemble methods such as Random Forest (RF), Gradient Boosting, and XGBoost [7, 11]. These architectures excel at capturing non-linear feature interactions. For example, ensemble models have established that superior ionic conductivity often correlates with an optimal degree of structural disorder, specific threshold values of sublattice bond ionicity, and optimal free volume per atom [7, 11]. Advanced ensemble stacking mechanisms synthesize multiple regressors to achieve extremely high accuracy—often exceeding 95%—in predicting the ionic conductivity of varied solid

electrolytes from complex feature sets [6, 7].

Beyond conductivity, mechanical and electrochemical limits dictate the ultimate viability of an SSB. Machine learning models have been successfully trained to directly predict bulk and shear moduli from lattice configurations. By screening tens of thousands of inorganic solids, these algorithms can identify soft, highly anisotropic materials that maintain excellent interfacial physical contact with the anode while simultaneously possessing the requisite shear strength to mechanically suppress lithium dendrite initiation [11, 43]. Similarly, data-driven regression models evaluating the elemental composition of solvents and solids isolate critical dependencies, such as the direct correlation between reduced solvent oxygen content and superior Coulombic efficiency in lithium metal batteries [11, 12].

Graph Neural Networks (GNNs)

Traditional grid-based and feature-engineered models suffer from substantial information loss when compressing complex 3D crystals into fixed-length 1D vectors. Graph Neural Networks (GNNs) preserve this fundamental topological information by mathematically mapping atoms to nodes (encoded with elemental properties) and interatomic bonds to edges (encoded with spatial distances) [16, 17]. The Crystal Graph Convolutional Neural Network (CGCNN) established the baseline for this methodology, utilizing iterative message-passing layers where nodes continuously aggregate feature vectors from their local chemical neighborhoods [16]. Through successive pooling layers, the network generates a holistic, highly dimensional representation of the crystal, predicting properties like formation energy, Fermi level, and band gaps with meV-level precision [16, 17].

The predictive capacity of GNNs has evolved significantly through architectural refinements, most notably the Atomistic Line Graph Neural Network (ALIGNN). While baseline GNNs encode only two-body interactions (node-to-node distances), ALIGNN constructs an auxiliary "line graph" where the original bonds become nodes, and the bond angles become edges [18]. This dual-graph architecture explicitly encodes three-body angular correlations. In solid-state electrolytes, where precise geometric polyhedral distortions and bond angles dictate the electrostatic landscape of an ion's migration pathway, capturing these three-body interactions is critical [3, 18]. Consequently, angle-aware GNNs like ALIGNN reduce prediction errors for structurally sensitive properties by up to 85% relative to their distance-only predecessors [18]. Additional models, such as CraLiGNN, further enhance this framework by incorporating compositionally restricted atomistic representations, drastically improving the prediction of thermal and electrical transport properties [18].

Active Learning and Bayesian Optimization

To maximize the efficiency of computationally expensive *ab initio* validations, static high-throughput screening is frequently superseded by Active Learning and Bayesian Optimization (BO). BO models the potential energy surface or the global material property landscape using probabilistic surrogate models, typically Gaussian Processes (GPs) [12, 13]. GPs are highly advantageous because they yield both a predicted mean property value and a quantifiable uncertainty variance for every hypothetical candidate in the chemical space [12]. In the context of SSE discovery, an acquisition function balances exploitation (sampling

materials structurally and chemically similar to known superionic conductors) and exploration (sampling highly uncertain, untested chemical families) [7, 12]. This iterative feedback loop dramatically accelerates the identification of anomalous outliers. For instance, BO has been

instrumental in discovering novel halogen-rich electrolytes (e.g., Li_3YBr_6) by optimizing the trade-off between the lithium diffusion barrier and the electronic band gap [12]. It is also utilized to explore the vast phase space of multi-anion composites, identifying optimal dopant ratios that maximize the volume of low-energy diffusion pathways while retaining thermodynamic stability [7, 12].

Methods: Machine Learning Interatomic Potentials (MLIPs)

The most transformative application of artificial intelligence in atomistic materials modeling is the advent of Machine Learning Interatomic Potentials (MLIPs). While static screening identifies promising structures, *ab initio* molecular dynamics (AIMD) simulations are imperative for observing the precise dynamic mechanisms of ion transport, concerted multi-ion hopping, and phase transitions at operational temperatures [3, 5, 29]. However, DFT-based AIMD is notoriously restricted by computational complexity, limiting simulations to picosecond

timescales and roughly 10^2 to 10^3 atoms. MLIPs bypass the explicit computation of electronic wavefunctions by mapping the local atomic environment directly to the potential energy surface (PES), yielding atomic energies, forces, and stress tensors. This paradigm enables molecular dynamics simulations at the nanosecond or microsecond scale for millions

of atoms—scaling linearly ($O(N)$) with system size while retaining near-quantum accuracy [22, 23, 28, 29, 30].

Architectural Evolution of MLIPs

Early MLIPs, such as the Behler-Parrinello High-Dimensional Neural Network Potentials (HDNNPs), Gaussian Approximation Potentials (GAPs), and Moment Tensor Potentials (MTPs), relied on fixed, hand-crafted descriptors to ensure rotational and translational invariance of the molecular system [22, 23, 24]. While highly accurate for specific, narrowly defined systems, their generalizability across diverse chemical spaces was severely restricted.

The current state-of-the-art leverages Equivariant Message Passing Neural Networks (MPNNs). Rather than utilizing invariant scalar descriptors that lose directional information, equivariant networks utilize tensor representations—employing spherical harmonics and Clebsch-Gordan coefficients—that predictably rotate in tandem with the 3D coordinate system [25, 26]. Models such as MACE (Message-passing Atomic Cluster Expansion) expand the body-order of message passing. By explicitly evaluating four-body correlations within a single message-passing layer, MACE reduces the required number of iterative network layers to just two. This architectural efficiency dramatically accelerates inference speed and enables massive GPU parallelization for large-scale solid-state simulations in software environments like LAMMPS [26, 27].

Universal Foundation Models

A profound paradigm shift has occurred with the pretraining of "universal" MLIPs (uMLIPs) on massive aggregated datasets spanning the entire periodic table. These foundation models are typically trained on tens of millions of structural relaxation trajectories sourced from the Materials Project and analogous databases [19, 20, 21]. By learning fundamental physical chemistry rules from vast data, these foundation models generalize across disparate chemical spaces without requiring *from-scratch* training for each new electrolyte system.

1. **CHGNet (Crystal Hamiltonian Graph Neural Network):** Distinctively, CHGNet incorporates site-specific magnetic moments into its training loop alongside standard energies and forces [19]. This charge-informed architecture enables the model to implicitly learn the orbital occupancy of electrons. In SSB materials, where redox activity at the electrode-electrolyte interface triggers heterogeneous charge redistribution and dynamic defect formation, tracking charge states is vital. CHGNet successfully captures these complex charge transfers during dynamic simulations, such as lithium insertion into transition metal oxides, making it indispensable for modeling battery degradation [19].
2. **M3GNet:** Leveraging a 3-body graph network framework, M3GNet encodes complex local chemistries and enables universal property prediction and zero-shot relaxation of arbitrary periodic crystals [20]. It has demonstrated extreme utility in filtering millions of hypothetical structures to find low-energy polymorphs, establishing itself as a robust surrogate for DFT geometry optimizations across the periodic table [20].
3. **MatterSim & SevenNet:** MatterSim utilizes deep graph neural networks trained via active learning paradigms applied to wide thermodynamic regimes (spanning temperatures from 0 to 5000 K and pressures up to 1000 GPa). It demonstrates exceptional proficiency in predicting Gibbs free energies and extreme-condition dynamics with minimal data requirements [30]. Similarly, SevenNet employs a knowledge distillation framework, where a massive multi-task teacher model transfers highly generalized representations to a compact, computationally efficient "SevenNet-Nano" model. This yields order-of-magnitude speedups for nanoscale molecular dynamics without compromising sub-angstrom interaction fidelity, making it highly suitable for large-scale liquid electrolyte and solid interface simulations [28].
4. **Orb (Orb-v3):** The Orb models employ sophisticated non-conservative force predictions and highly flexible structural representations. In exhaustive comparative benchmarks analyzing lithium solid ion conductors, Orb-v3 consistently demonstrated state-of-the-art accuracy in predicting static migration barriers via NEB calculations. It correctly classifies good versus poor ionic conductors with extreme precision, proving its utility for the zero-shot extraction of kinetic transport properties [31, 32].

Universal MLIP	Core Architectural Feature	Key SSB Application Strength
MACE	Equivariant, higher body-order (4-body)	Highly parallelizable MD; fast evaluation of highly

	messages	disordered solid electrolytes [26, 27].
CHGNet	Pretrained on magnetic moments / charge states	Simulating redox reactions and charge transfers at solid-solid interfaces [19].
M3GNet	Explicit 3-body angular interaction mapping	Universal structural relaxation and phase stability prediction across the periodic table [20].
SevenNet	Knowledge distillation to compact GNNs	Extreme computational efficiency for multi-million atom simulations; liquid electrolyte dynamics [28].
MatterSim	Active learning across massive P/T phase spaces	Extreme condition thermodynamics; finite-temperature phase diagram prediction [30].
Orb-v3	Flexible non-conservative force predictions	State-of-the-art prediction of static migration barriers and dynamic ion transport kinetics [31, 32].

Data-Efficient Fine-Tuning and Continual Learning

Despite the unprecedented breadth of universal MLIPs, zero-shot predictions can occasionally falter when applied to highly out-of-distribution dynamic phenomena, such as complex interfacial degradation mechanisms or high-temperature phase transitions. Re-training foundation models from scratch on specialized datasets is computationally exorbitant. Thus, advanced fine-tuning strategies have been formulated to adapt these models.

The FIRE (Fine-tuning with Integrated Replay and Efficiency) framework represents a significant methodological leap in this domain [34, 35]. When neural networks are fine-tuned on a narrow, highly specific dataset (e.g., a specific solid-solid interface), they frequently suffer from "catastrophic forgetting," losing their generalized understanding of basic physical chemistry learned during pretraining. By utilizing a replay-augmented continual learning strategy, FIRE interleaves task-specific interface data with subsampled anchor data from the original

pretraining distribution [34]. Applied to solid-solid battery interfaces, FIRE has demonstrated the ability to lower energy root-mean-square errors to sub-1 meV/atom using only a fraction of traditional data requirements, establishing a highly scalable pathway for precision atomistic modeling [34, 35].

Methods: Generative Artificial Intelligence for Crystal Structure Prediction

While discriminative high-throughput screening evaluates predetermined materials curated from historical databases, Generative AI inverts the scientific paradigm. It navigates the continuous, highly dimensional mathematical latent space of materials to predict entirely novel stoichiometries and crystal topologies *de novo*. Crystal Structure Prediction (CSP) is notoriously complex; unlike 1D text strings or 2D molecular graphs, inorganic solids possess 3D periodic boundary conditions, variable unit cell volumes, and must strictly adhere to one of the 230 space group symmetries [36, 37].

Diffusion Models and Variational Autoencoders

The Crystal Diffusion Variational Autoencoder (CDVAE) provides a mathematically rigorous approach to the CSP challenge [36]. Traditional generation methods often output unphysical structures with overlapping atoms or highly unstable lattices. CDVAE circumvents this by projecting known materials into an SE(3)-equivariant latent space. During the decoding phase, a score-matching diffusion process utilizes Langevin dynamics to systematically denoise fractional atomic coordinates, effectively simulating a thermodynamic relaxation toward a local energy minimum [36, 37, 38]. By implicitly learning the physical inductive biases of atomic bonding and quantum mechanics from the training data, CDVAE generates highly realistic, symmetry-aware materials. Conditional variants (Cond-CDVAE) elevate this capability by allowing researchers to explicitly input target physical parameters—such as desired activation energies, band gaps, or target pressure/temperature conditions—and directly sample crystal structures mathematically optimized for those specific characteristics [36, 38].

This diffusion-based structure prediction has yielded practical, groundbreaking discoveries in battery research. For example, CDVAE techniques were utilized to computationally generate

and validate amorphous phases of lithium difluorophosphate (LiPO_2F_2), a critical component of solid-electrolyte interphases. The generative models, coupled with MLIP

evaluations, proved that while the crystalline state of LiPO_2F_2 is an ionic insulator, the structural disorder inherent in the generated amorphous phase drastically flattens the local lithium site-energy landscape [36, 38]. This structural rearrangement results in an order-of-magnitude leap in intrinsic conductivity, fundamentally reshaping how the battery industry approaches the engineering of minority grain boundary components and disordered solid electrolytes.

GNoME and Multi-modal Generation

The unprecedented scalability of graph-based generative networks was vividly demonstrated

by the Graph Networks for Materials Exploration (GNoME) project [21]. By orchestrating a massive active learning pipeline that iteratively generated hypothetical crystals and validated them via autonomous DFT calculations, GNoME effectively expanded the convex hull of known thermodynamic stability for humanity. The framework generated over 2.2 million new crystals, identifying roughly 380,000 highly stable materials. Among these, researchers identified hundreds of novel lithium-ion conductors that had never been previously theorized [21]. The implication of such sheer combinatorial scale is profound: the primary bottleneck in energy materials science has been definitively shifted from *virtual discovery* to *experimental synthesizability*.

Furthermore, the integration of Large Language Models (LLMs) with crystal generation is rapidly expanding predictive boundaries. By fine-tuning foundational LLMs (such as LLaMA-2 or Mistral) on vast corpora of materials science literature and textual chemical descriptions alongside topological data, researchers are developing multi-modal generation strategies. Dual-AI approaches—pairing CDVAE for structural variation with finely tuned LLMs for enforcing thermodynamic stability constraints—have rapidly identified previously unknown porous transition-metal oxides uniquely suited for advanced multivalent-ion conduction [21, 37, 38].

Interface Modeling and Closed-Loop Autonomous Synthesis

A solid-state battery is functionally limited not merely by the bulk conductivity of the SSE, but heavily by the kinetic and chemical friction occurring at its boundaries. Machine learning actively bridges the gap between bulk theory and operational reality by decoding complex interfaces and automating physical synthesis.

Decoding the Solid Electrolyte Interphase (SEI) and Grain Boundaries

In practical SSBs, the solid-solid interface between the electrolyte and the highly reactive lithium metal anode dynamically decomposes to form the Solid Electrolyte Interphase (SEI) [3, 41]. An optimal SEI must be electronically insulating to prevent continuous, parasitic electrolyte degradation, yet highly ionically conductive to allow efficient charge transfer during cycling [41]. The structural heterogeneity, extreme thinness, and amorphous nature of the SEI make it virtually impossible to model fully via static DFT calculations.

By employing large-scale MLIPs, researchers can now simulate nanosecond, multi-million-atom molecular dynamics trajectories showing the *in situ* formation and evolution of the SEI. Machine learning models reveal how specific localized electric fields at the grain boundaries of solid electrolytes enhance ionic resistance. This resistance prompts the build-up of localized electron leakage, which directly catalyzes the reduction of lithium ions and the consequent nucleation of destructive lithium metal dendrites [3, 11, 41]. ML-accelerated dynamics have successfully predicted how engineered chemical mechanisms—such as halide segregation, where chlorine or bromine atoms migrate to the interface during cycling—inhibit dendrite penetration by modulating local mechanical shear moduli [35, 41]. Dynamic voltage stability algorithms utilize high-throughput calculation methodologies to identify optimal buffer

coatings that decouple thermodynamic instability from mechanical failure, extending the cycle life of solid-state systems [34, 35].

Autonomous Laboratories and Solid-State Synthesis

The overwhelming volume of virtual predictions generated by MLIPs, GNoME, and CDVAEs demands an equivalent acceleration in physical experimental validation. Autonomous, AI-driven laboratories, prominently exemplified by the A-Lab, orchestrate this critical translation from the digital realm to physical reality [39].

Unlike solution-based organic synthesis, solid-state inorganic synthesis requires the complex handling of rigid powders, precise stoichiometric milling, and highly specific high-temperature calcination profiles. The A-Lab addresses these hardware challenges by employing natural language processing (NLP) models trained on decades of materials science literature to intelligently draft initial synthesis recipes, selecting optimal precursor materials and requisite heating profiles [39]. Robotic systems autonomously execute the dosing, mixing, and baking of these powders, followed by *in situ* X-ray diffraction (XRD) characterization of the output. An active learning algorithm analyzes the XRD spectra to detect phase purity and crystallographic structure. If the synthesis fails or results in undesirable multiphase mixtures, the system utilizes Bayesian logic to adjust the thermal profiles or stoichiometric ratios in real time for subsequent robotic iterations [39]. Over continuous 17-day operational periods, the A-Lab has proven capable of synthesizing 41 out of 58 highly complex, computationally predicted novel target compounds, demonstrating a success rate that far exceeds traditional human-driven trial and error and solidifying the feasibility of closed-loop materials discovery [39].

Challenges and Prospects

While the integration of machine learning has irrevocably altered the trajectory of solid-state battery research, the domain faces several substantive hurdles that define the immediate scientific frontier.

First, an overwhelming preponderance of existing databases, extraction methodologies, and pretrained foundation models are implicitly biased toward monovalent lithium-ion systems [5, 40]. The pursuit of multivalent solid-state architectures—utilizing earth-abundant ions such as Mg^{2+} , Ca^{2+} , Zn^{2+} , or Al^{3+} —is severely hampered by extreme data scarcity [40].

Multivalent ions exhibit vastly different charge densities and interact via much stronger electrostatic polarization mechanics within solid lattices. Consequently, MLIPs and GNNs explicitly pretrained on lithium trajectories exhibit high epistemic uncertainty and severe predictive degradation when transferred to multivalent domains. Future efforts require aggressive dataset augmentation, targeted active learning campaigns tailored specifically for multivalent mobility, and the broad implementation of transfer learning to bootstrap multivalent models from lithium-rich priors [31, 40].

Second, simulating far-from-equilibrium phenomena remains technically fraught. Battery degradation, extreme thermal runaway events, and high-voltage electrochemical phase transitions occur in thermodynamic states far from the stable ground state [31, 33]. Most foundational universal potentials (such as M3GNet or early iterations of MACE) are

predominantly trained on near-equilibrium structures relaxing toward 0 K energy minima [26, 31]. Utilizing these models to simulate high-rate charging protocols, fracture mechanics, or extreme defect concentrations requires rigorous uncertainty quantification. Techniques that continuously track per-atom uncertainty variance during inference are becoming mandatory to prevent ML-driven molecular dynamics simulations from entering unphysical, wildly extrapolated regimes [32, 33].

Finally, the interpretability of deep learning models must be rigorously elevated. Although techniques like SHAP (SHapley Additive exPlanations) values and attention weights provide some insight into which geometric features drive conductivity predictions, extracting actionable, generalizable physical laws—*physics-informed AI*—remains an ongoing challenge [37, 42]. Ensuring that generative models produce phases that are not only thermodynamically viable but also economically manufacturable from accessible precursors is critical [39]. The bridging of generative AI with realistic processability constraints, guided by platforms like the A-Lab, signifies the necessary evolution from merely identifying theoretical superionic conductors to engineering commercial, deployable solid-state batteries [39].

Conclusion

The deployment of machine learning in solid-state battery materials discovery has rapidly matured from isolated theoretical demonstrations to a foundational pillar of modern computational chemistry and energy science. By effectively mitigating the combinatorial explosion of the inorganic chemical space, artificial intelligence provides a streamlined, predictive conduit spanning fundamental electron interactions to macroscopic device behavior. The curation of extensive thermodynamic databases coupled with the predictive fidelity of topological and composition-aware graph neural networks has enabled the instantaneous, high-throughput screening of millions of hypothetical electrolytes, identifying outliers with optimal combinations of ionic conductivity, stability, and mechanical strength.

Concurrently, the deployment of equivariant, universal machine learning interatomic potentials—such as MACE, CHGNet, M3GNet, and Orb—successfully bridges the quantum-mechanical accuracy of density functional theory with the spatiotemporal requirements of large-scale molecular dynamics. These foundation models demystify critical kinetic phenomena that were previously computationally inaccessible, including interfacial degradation, continuous defect formation, and dynamic ionic mobility at the nanoscale. Moreover, the transition from discriminative screening to generative *de novo* design, championed by diffusion models like CDVAE and massive active learning frameworks like GNoME, ensures a continual supply of geometrically and physically valid structural candidates, pushing the boundaries of the known chemical universe.

The ultimate validation of this data-driven pipeline is increasingly realized through closed-loop autonomous laboratories, which harmonize virtual AI predictions with robotic synthesis and real-time active learning optimization. As methodologies iteratively evolve to confront prevailing challenges—such as mitigating severe data bias in multivalent systems, enhancing the fidelity of far-from-equilibrium simulations, and advancing physics-informed interpretability—the synergy of artificial intelligence and materials science stands uniquely

poised to actualize the next generation of safe, sustainable, and high-energy-density solid-state energy storage solutions.

References

- [1] L. Zhao, J. Liang, J. T. Kim, X. Sun, "Prospects of halide-based all-solid-state batteries: From material design to practical application," *Science Advances*, 8(36), eadc9516, 2022. DOI: 10.1126/sciadv.adc9516 [2] Q. Zhao, S. Stalin, C.-Z. Zhao, L. A. Archer, "Designing solid-state electrolytes for safe, energy-dense batteries," *Nature Reviews Materials*, 5(3), 229-252, 2020. DOI: 10.1038/s41578-019-0165-5 [3] A. C. C. Dutra, B. A. Goldmann, M. S. Islam, J. A. Dawson, "Understanding solid-state battery electrolytes using atomistic modelling and machine learning," *Nature Reviews Materials*, 11(2), 120-142, 2026. DOI: 10.1038/s41578-025-00842-1 [4] S. Wang, L. Yu, L. Zhang, X. Sun, "Artificial Intelligence Empowers Solid-State Batteries for Material Screening and Performance Evaluation," *Nano-Micro Letters*, 16(1), 112-138, 2024. DOI: 10.1007/s40820-023-01254-x [5] H. H. Guo, Q. Wang, A. Stuke, A. Urban, "Accelerated Atomistic Modeling of Solid-State Battery Materials With Machine Learning," *Frontiers in Energy Research*, 9, 695902, 2021. DOI: 10.3389/fenrg.2021.695902 [6] X. Li, Y. Zhang, M. S. Islam, J. A. Dawson, "Machine Learning-Assisted Property Prediction of Solid-State Electrolyte," *Advanced Energy Materials*, 14(12), 2304480, 2024. DOI: 10.1002/aenm.202304480 [7] M. Ait Tamer, X. Lin, X. Sun, A. Lahmar, J. Ma, M. Yang, "Accelerated Discovery of Solid-State Battery Properties Enabled by Active Learning Approaches," *Journal of Materials Chemistry A*, 12, 8452-8463, 2024. DOI: 10.1039/D3TA07245J [8] A. Sendek, E. D. Cubuk, E. J. Reed, "Accelerating the design and discovery of solid lithium ion conductor materials with physics-based machine learning," *Chemistry of Materials*, 29(18), 7982-7993, 2017. DOI: 10.1021/acs.chemmater.7b02352 [9] A. Sendek, Q. Yang, E. D. Cubuk, E. J. Reed, "Machine Learning-Assisted Discovery of Solid Li-Ion Conducting Materials," *Chemistry of Materials*, 31(2), 342-352, 2019. DOI: 10.1021/acs.chemmater.8b03272 [10] E. D. Cubuk, A. Sendek, E. J. Reed, "Screening of Lithium Solid-State Electrolytes Through Machine Learning," *Frontiers in Energy Research*, 9, 639741, 2021. DOI: 10.3389/fenrg.2021.639741 [11] Z. Ahmad, T. Viswanathan, C. Wang, V. Viswanathan, "Machine Learning Enabled Computational Screening of Inorganic Solid Electrolytes for Suppression of Dendrite Formation in Lithium Metal Anodes," *ACS Applied Materials & Interfaces*, 10(31), 26134-26141, 2018. DOI: 10.1021/acsami.8b02352 [12] S. A. Tawfik, J. Berk, T. R. Walsh, S. Rana, S. Venkatesh, "Accelerated Discovery of Solid-State Electrolytes Using Bayesian Optimization," *The Journal of Physical Chemistry C*, 129(8), 4523-4534, 2025. arXiv: 2410.02643. DOI: 10.1021/acs.jpcc.5c00954 [13] G. Hargreaves, L. Laskowski, M. J. Rosseinsky, "OBELiX: A Curated Dataset of Crystal Structures and Experimentally Measured Ionic Conductivities for Lithium Solid-State Electrolytes," arXiv:2502.14234, 2025. [14] B. Chen, S. P. Ong, "Dynamic Database of Solid-State Electrolyte (DDSE) for accelerated material discovery," *Computational Materials Science*, 234, 112830, 2024. DOI: 10.1016/j.commatsci.2024.112830 [15] L. Laskowski, G. Hargreaves, M. J. Rosseinsky, "The Liverpool Ionics (Lilong) Dataset for solid-state battery electrolytes," *Scientific Data*, 10, 432, 2023. DOI: 10.1038/s41597-023-02341-z [16] T. Xie, J. C. Grossman, "Crystal Graph Convolutional Neural Networks for an Accurate and Interpretable Prediction of Material Properties," *Physical Review Letters*, 120(14), 145301, 2018. arXiv: 1710.10324. DOI: 10.1103/PhysRevLett.120.145301 [17] C. Chen, W. Ye, Y. Zuo, C. Zheng, S. P.

Ong, "Graph Networks as a Universal Machine Learning Framework for Molecules and Crystals," *Chemistry of Materials*, 31(9), 3564–3561, 2019. arXiv: 1812.05055. DOI: 10.1021/acs.chemmater.9b01294 [18] K. Choudhary, B. DeCost, "Atomistic Line Graph Neural Networks for enhanced materials property predictions," *npj Computational Materials*, 7, 185, 2021. arXiv: 2106.01254. DOI: 10.1038/s41524-021-00650-1 [19] B. Deng, P. Zhong, K. J. Jun, J. Riebesell, K. Han, C. J. Bartel, G. Ceder, "CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling," *Nature Machine Intelligence*, 5(9), 1031–1041, 2023. arXiv: 2302.14231. DOI: 10.1038/s42256-023-00716-3 [20] C. Chen, S. P. Ong, "A universal graph deep learning interatomic potential for the periodic table," *Nature Computational Science*, 2(11), 718–728, 2022. arXiv: 2202.02450. DOI: 10.1038/s43588-022-00349-3 [21] A. Merchant, S. Batzner, S. S. Schoenholz, S. P. Ong, E. D. Cubuk, "Millions of new materials discovered with deep learning," *Nature*, 624, 80–85, 2023. DOI: 10.1038/s41586-023-06735-9 [22] J. Behler, M. Parrinello, "Generalized Neural-Network Representation of High-Dimensional Potential-Energy Surfaces," *Physical Review Letters*, 98(14), 146401, 2007. DOI: 10.1103/PhysRevLett.98.146401 [23] A. P. Bartók, M. C. Payne, R. Kondor, G. Csányi, "Gaussian Approximation Potentials: The Accuracy of Quantum Mechanics, without the Electrons," *Physical Review Letters*, 104(13), 136403, 2010. arXiv: 0910.1019. DOI: 10.1103/PhysRevLett.104.136403 [24] A. V. Shapeev, "Moment Tensor Potentials: A Class of Systematically Improvable Interatomic Potentials," *Multiscale Modeling & Simulation*, 14(3), 1153–1173, 2016. arXiv: 1512.06054. DOI: 10.1137/15M1054183 [25] S. Batzner, A. Musaelian, L. Sun, M. Geiger, J. P. Mailoa, M. Kornbluth, N. Feola, S. T. Simon, B. Kozinsky, "E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials," *Nature Communications*, 13, 2453, 2022. arXiv: 2101.03164. DOI: 10.1038/s41467-022-29939-5 [26] I. Batatia, D. P. Kovács, G. N. C. Simm, C. Ortner, G. Csányi, "MACE: Higher Order Equivariant Message Passing Neural Networks for Fast and Accurate Force Fields," *Advances in Neural Information Processing Systems*, 35, 11423–11436, 2022. arXiv: 2206.07697. [27] I. Batatia, S. Batzner, D. P. Kovács, A. Musaelian, G. N. C. Simm, C. Ortner, B. Kozinsky, G. Csányi, "A foundation model for atomistic materials chemistry," arXiv:2401.00096, 2024. [28] Y. Park, J. Kim, S. Hwang, S. Han, "Scalable Parallel Algorithm for Graph Neural Network Interatomic Potentials in Molecular Dynamics Simulations," *Journal of Chemical Theory and Computation*, 20(11), 4857–4868, 2024. DOI: 10.1021/acs.jctc.4c00190 [29] J. Kim, J. Kim, J. Kim, J. Lee, Y. Park, Y. Kang, S. Han, "Data-Efficient Multifidelity Training for High-Fidelity Machine Learning Interatomic Potentials," *Journal of the American Chemical Society*, 147(1), 1042–1054, 2024. DOI: 10.1021/jacs.4c14455 [30] G. Yang, T. Tahmasbi, Y. Xu, L. Zhang, et al., "MatterSim: A Deep Learning Atomistic Model Across Elements, Temperatures and Pressures," arXiv:2405.04967, 2024. [31] H. Du, J. Hui, L. Zhang, H. Wang, "Universal Machine Learning Interatomic Potentials are Ready for Solid Ion Conductors," arXiv:2502.09970, 2025. [32] H. Du, J. Hui, Y. Zhou, "Assessment and Application of Universal Machine Learning Interatomic Potentials in Solid-State Electrolyte Research," *ACS Materials Letters*, 7, 1123–1135, 2025. DOI: 10.1021/acsmaterialslett.5c00336 [33] N. Alghamdi, et al., "Comparing fine-tuning strategies for machine learning force fields in lithium-ion diffusion," arXiv:2510.05020, 2025. [34] X. Liu, X. Yu, et al., "An AI-ready fine-tuning framework for accurate machine-learning interatomic potentials in solid-solid battery interfaces," arXiv:2601.17847, 2026. [35] K. Kim, S. Yuan, B. C. Wood, L. F.

Wan, "Machine-learning interatomic potentials for interfaces in all-solid-state batteries: Perspectives on training data, model selection, and validation," *MRS Communications*, 16, 145-156, 2026. DOI: 10.1557/s43579-026-00928-9 [36] T. Xie, X. Fu, O.-E. Ganea, R. Barzilay, T. Jaakkola, "Crystal Diffusion Variational Autoencoder for Periodic Material Generation," *International Conference on Learning Representations (ICLR)*, 2022. arXiv: 2110.06197. [37] A. Vasylenko, F. Ottomano, C. M. Collins, R. Savani, M. S. Dyer, M. J. Rosseinsky, "Introducing physics-informed generative models for targeting structural novelty in the exploration of chemical space," arXiv:2510.23181, 2025. [38] C. M. Collins, H. M. Sayeed, G. R. Darling, J. B. Claridge, T. D. Sparks, M. J. Rosseinsky, "Integration of generative machine learning with the heuristic crystal structure prediction code FUSE," *Faraday Discussions*, 256, 85-103, 2024. DOI: 10.1039/D4FD00094C [39] N. J. Szymanski, B. Rendy, Y. Fei, R. E. Kumar, T. He, D. Milsted, M. J. McDermott, M. Gallant, E. D. Cubuk, A. Merchant, H. Kim, A. Jain, C. J. Bartel, K. Persson, Y. Zeng, G. Ceder, "An autonomous laboratory for the accelerated synthesis of novel materials," *Nature*, 624, 414-418, 2023. DOI: 10.1038/s41586-023-06734-w [40] L. Zhao, J. Liang, J. T. Kim, X. Sun, "Prospects of halide-based all-solid-state batteries: From material design to practical application," *Science Advances*, 8(36), eadc9516, 2022. DOI: 10.1126/sciadv.adc9516 [41] A. Aouichaoui, et al., "Machine Learning Potentials for Electrified Interfaces," *ChemRxiv*, 2026. [42] H. Kaur, et al., "Data-efficient fine-tuning of foundational models for first-principles...", *Faraday Discussions*, 2025. [43] L. Y. A. Choong, "Halide solid electrolytes...", *ACS Applied Energy Materials*, 2025.